

Performance Marketing Formula & Diagnostic Framework

Master mathematical dependencies, business unit economics, and operational ad channel flows.

Core Performance Metrics

FOUNDATION

$$\text{CTR} = (\text{Clicks} \div \text{Impressions}) \times 100$$

$$\text{CPC} = \text{Spend} \div \text{Clicks}$$

$$\text{CVR} = (\text{Conversions} \div \text{Clicks}) \times 100$$

$$\text{CPA} = \text{Spend} \div \text{Conversions}$$

$$\text{CPL} = \text{Spend} \div \text{Leads}$$

$$\text{CPM} = (\text{Spend} \div \text{Impressions}) \times 1,000$$

$$\text{ROAS} = \text{Revenue} \div \text{Spend}$$

$$\text{AOV} = \text{Revenue} \div \text{Orders}$$

Key Inter-Metric Relationships

THE POWER RATIOS

$$\text{CPA} = \text{CPC} \div \text{CVR}$$

$$\text{ROAS} = \text{AOV} \div \text{CPA}$$

- CPA is strictly governed by incoming traffic cost (CPC) vs on-site efficiency (CVR).
- ROAS scales by elevating basket size (AOV) or reducing acquisition cost (CPA).

Meta Specific Metrics

AUCTION & ADS

$$\text{Frequency} = \text{Impressions} \div \text{Reach}$$

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$$\text{Link CTR} = (\text{Link Clicks} \div \text{Impressions}) \times 100$$

$$\text{Outbound CTR} = (\text{Outbound Clicks} \div \text{Impressions}) \times 100$$

$$\text{Cost per Result} = \text{Spend} \div \text{Results}$$

Business Economics & Profitability

FINANCIAL VIABILITY

$$\text{COGS} = \text{Cost of Goods Sold}$$

$$\text{Gross Profit} = \text{Revenue} - \text{COGS}$$

$$\text{Gross Margin \%} = (\text{Gross Profit} \div \text{Revenue}) \times 100$$

$$\text{Break-even ROAS} = 1 \div \text{Gross Margin \%}$$

$$\text{Contribution Profit} = \text{Revenue} - \text{COGS} - \text{Ad Spend} - \text{Variable Costs}$$

- Target ROAS must exceed Break-even ROAS for positive net contribution cash flow.

Customer Acquisition Economics

BUSINESS ACQUISITION

$$\text{CAC} = \text{Total Acquisition Cost} \div \text{New Customers}$$

$$\text{Marginal CAC} = \text{Incremental Spend} \div \text{Incremental Customers}$$

$$\text{Marginal ROAS} = \text{Incremental Revenue} \div \text{Incremental Spend}$$

$$\text{If AOV is stable: Marginal ROAS} \approx \text{AOV} \div \text{Marginal CAC}$$

- **CAC vs CPA / ROAS:** CPA tracks conversions; CAC tracks new buyers. ROAS measures revenue efficiency; CAC measures customer efficiency.
- **Scaling Principle:** Don't judge scale solely by average CPA/ROAS. Watch Marginal CAC and Marginal ROAS for incremental viability.

Channel Mental Flows (Linear Diagnostic Chains)

END-TO-END REASONING

Mental Flow — Google Ads (Search & High Intent)

Impressions → (filtered by) **CTR** → yields **Clicks** → at a unit cost of **CPC** → (converted on LP via) **CVR** → producing **Conversions** → evaluated against unit acquisition cost **CPA** ($\text{CPC} \div \text{CVR}$) → generating **Revenue / AOV** → calculating ad efficiency **ROAS** ($\text{AOV} \div \text{CPA}$) → subtracted by **COGS** → determining product **Gross Margin %** → establishing **Break-even ROAS** → resulting in net cash flow **Contribution / Profit**.

Mental Flow — Meta Ads (Social & Interruption Discovery)

Reach → (multiplied by repetition) **Frequency** → generates total **Impressions** → filtered by hook/creative via **CTR / Link CTR / Outbound CTR** → driving outbound **Clicks** → purchased at auction unit rate **CPC** → converted through funnel via **CVR** → completing **Conversions / Results** → measured at unit acquisition rate **CPA / Cost per Result** → monetized via **Revenue / AOV** → determining scaling efficiency **ROAS** → factoring manufacturing / sourcing **COGS** → defining commercial **Gross Margin %** → benchmarked against **Break-even ROAS** → closing at net financial **Contribution / Profit**.

Upper Funnel & Traffic Diagnostics

Golden Rule: Observe Movement → Use Formula → Identify Possible Drivers → Investigate Drivers.

1. Impressions Changes

AUCTION & SCALE

▼ IMPRESSIONS DECREASED (↓) — CHECK:

- **Budget / Spend:** Daily caps hit, pacing throttled, or billing paused.
- **Delivery & Bidding:** Target bid cap set too low; outbid in auction.
- **Targeting / Audience:** Overly tight segment, extreme exclusions, overlap.
- **Ad Eligibility:** Policy issues, review delay, ad quality score penalty.

▲ IMPRESSIONS INCREASED (↑) — CHECK:

- Budget increased, audience broadened, or auction competition became favorable.

2. CTR (Click-Through Rate)

CREATIVE RESONANCE

$$\text{CTR} = (\text{Clicks} \div \text{Impressions}) \times 100$$

▼ CTR DECREASED (↓) — CHECK:

- **Creative Fatigue:** Ad saturated, audience exhausted with old hook.
- **Ad Relevance:** Weak value hook, generic visuals, audience mismatch.
- **Audience Fit:** Algorithms pushing into broader, unengaged cohorts.
- **Placements:** Inventory shifting into passive or low-intent feeds.

▲ CTR INCREASED (↑) — CHECK:

- Fresh winning creative, tighter message-market fit, or high-intent placements.

3. Clicks Changes

TRAFFIC VOLUME

$$\text{Clicks} = \text{Impressions} \times \text{CTR}$$

▼ CLICKS DECREASED (↓) — CHECK:

- **Formula Drivers:** Impressions ↓ OR CTR ↓ (or both falling together).
- Determine whether it is an *auction volume problem* (Impressions) or an *engagement problem* (CTR).

▲ CLICKS INCREASED (↑) — CHECK:

- **Formula Drivers:** Impressions ↑ OR CTR ↑ (or both). Cross-check LP bounce rate to ensure real visitor quality.

4. CPC (Cost Per Click)

CLICK EFFICIENCY

$$\text{CPC} = \text{Spend} \div \text{Clicks}$$

▲ CPC INCREASED (↑) — CHECK:

- **Direct Drivers:** Spend ↑ OR Clicks ↓ (or both).
- **Drill Down:** If Clicks ↓ → is CTR ↓ (fatigue) or Impressions ↓ (auction)?
- Examine rising CPM auction costs, aggressive manual bids, or degraded relevance.

▼ CPC DECREASED (↓) — CHECK:

- Spend ↓ OR Clicks ↑. Favorable bidding auction, cheaper placements, or surging CTR.

5. CVR (Conversion Rate)

LANDING PAGE / FUNNEL

$$\text{CVR} = (\text{Conversions} \div \text{Clicks}) \times 100$$

▼ CVR DECREASED (↓) — CHECK:

- **Traffic Intent:** Irrelevant keyword searches or misleading ad message.
- **Landing Page UX:** Slow page load, broken mobile layout, form bugs.
- **Offer Friction:** Pricing friction, sudden shipping fees, lack of social proof.
- **Tracking / Tech:** Payment gateway failure, broken scripts, cookie tag loss.

▲ CVR INCREASED (↑) — CHECK:

- Conversions ↑ OR Clicks ↓. High intent buyers, streamlined 1-page checkout, clear offer.

6. Conversions Changes

OUTCOME VOLUME

$$\text{Conversions} = \text{Clicks} \times \text{CVR}$$

▼ CONVERSIONS DECREASED (↓) — CHECK:

- **Formula Drivers:** Clicks ↓ OR CVR ↓ (or both).
- Identify whether the bottleneck sits in marketing traffic generation (Clicks) or on-site checkout funnel (CVR).

▲ CONVERSIONS INCREASED (↑) — CHECK:

- Clicks ↑ OR CVR ↑ (or both). Validate CPA & profit margins to maintain scaling health.

Cost, Lead & Auction Diagnostics

Deep dive into unit acquisition cost, lead capture, auction pressure, and basket sizes.

7. CPA (Cost Per Acquisition)

UNIT EFFICIENCY

$$\text{CPA} = \text{CPC} \div \text{CVR} = \text{Spend} \div \text{Conversions}$$

▲ CPA INCREASED (↑) — CHECK:

- **Core Equation:** CPC ↑ OR CVR ↓ (or both).
- **If CPC ↑:** Spend ↑ OR Clicks ↓ → Check auction bidding, fatigue, CPM inflation.
- **If CVR ↓:** Conversions ↓ OR Clicks ↑ → Check LP latency, checkout friction, tracking drop.

▼ CPA DECREASED (↓) — CHECK:

- CPC ↓ OR CVR ↑ (or both). Compounding efficiency gains across traffic & funnel.

8. CPL (Cost Per Lead)

LEAD EFFICIENCY

$$\text{CPL} = \text{Spend} \div \text{Leads}$$

▲ CPL INCREASED (↑) — CHECK:

- **Direct Drivers:** Spend ↑ OR Leads ↓ (or both).
- **Underlying Factors:** Rising CPCs, lower traffic, or drop in landing page form fills.
- **Quality Check:** Evaluate lead qualification fields vs form friction and bot spam.

▼ CPL DECREASED (↓) — CHECK:

- Spend ↓ OR Leads ↑. Streamlined forms, native lead ads, stronger value magnet.

9. CPM (Cost Per Mille)

AUCTION PRESSURE

$$\text{CPM} = (\text{Spend} \div \text{Impressions}) \times 1,000$$

▲ CPM INCREASED (↑) — CHECK:

- **Auction & Seasonality:** Q4/holiday spikes, major competitor aggressive bids.
- **Audience:** Highly constrained audience sizes, narrow geo-targeting, heavy overlap.
- **Placements & Quality:** Constrained to expensive inventory, low ad quality score.

▼ CPM DECREASED (↓) — CHECK:

- Spend ↓ OR Impressions ↑. Broader audience cohorts, off-peak delivery, placement liquidity.

10. AOV (Average Order Value)

BASKET ECONOMICS

$$\text{AOV} = \text{Revenue} \div \text{Orders}$$

▼ AOV DECREASED (↓) — CHECK:

- **Product Mix:** Traffic skewing heavily to low-ticket introductory items.
- **Pricing & Promos:** Excessive coupon discounting cannibalizing basket revenue.
- **Upsell Performance:** Broken post-purchase cross-sell or bundle modules.
- **Customer Cohort:** Increased share of bargain hunters or price-sensitive users.

▲ AOV INCREASED (↑) — CHECK:

- Revenue ↑ OR Orders ↓. Bundles, minimum spend free-shipping thresholds, upsells.

11. ROAS (Return On Ad Spend)

EFFICIENCY MULTIPLIER

$$\text{ROAS} = \text{Revenue} \div \text{Spend} = \text{AOV} \div \text{CPA}$$

▼ ROAS DECREASED (↓) — CHECK:

- **Primary Ratio:** AOV ↓ OR CPA ↑ (or both).
- If **CPA ↑:** Drill down into CPC (CPM, CTR) and CVR (Landing Page/Offer).
- If **AOV ↓:** Investigate cart size decay or heavy promo code dilution.

▲ ROAS INCREASED (↑) — CHECK:

- AOV ↑ OR CPA ↓ (or both). Compounded gains from targeting and high-ticket baskets.

12. Revenue Changes

TOPLINE SCALE

$$\text{Revenue} = \text{Conversions} \times \text{AOV}$$

▼ REVENUE DECREASED (↓) — CHECK:

- **Formula Drivers:** Conversions ↓ OR AOV ↓ (or both).
- Check macro demand patterns, tracking outages, or high-velocity inventory stock-outs.

▲ REVENUE INCREASED (↑) — CHECK:

- Conversions ↑ OR AOV ↑ (or both). Successful channel scaling with healthy unit economics.

Delivery, Saturation & Profitability

Meta platform delivery mechanics, creative fatigue, and unit economics profitability.

13. Spend Changes

PACING & BUDGET

▲ SPEND INCREASED (↑) — CHECK:

- Manual budget increase or automated scaling rules triggered.
- Delivery expansion into higher CPM auctions or competitive bid pools.
- Bid cap loosening causing aggressive clearing bid prices.

▼ SPEND DECREASED (↓) — CHECK:

- Budget cuts, billing threshold pauses, ad disapproval, tight bid caps.

14. Frequency Changes (Meta)

SATURATION ALERT

$$\text{Frequency} = \text{Impressions} \div \text{Reach}$$

▲ FREQUENCY ↑ + CTR ↓ + CVR ↓ (CRITICAL ALERT):

- **Audience Saturation / Fatigue:** Same audience has repeatedly seen creatives.
- **Remedy:** Refresh creative angles immediately, widen audience, exclude past 30-day purchasers.

▼ FREQUENCY DECREASED (↓) — CHECK:

- Reach ↑ OR Impressions ↓. Broadening audience or scaling fresh cohorts.

15. Reach Changes (Meta)

UNIQUE EXPOSURE

$$\text{Reach} = \text{Impressions} \div \text{Frequency}$$

▼ REACH DECREASED (↓) — CHECK:

- Impressions ↓ OR Frequency ↑ (or both).
- Audience pool constrained by excessive exclusion filters or bid restrictions.
- Ad delivery trapped in narrow retargeting loops rather than top-of-funnel prospecting.

▲ REACH INCREASED (↑) — CHECK:

- Targeting widened, lookalike expansion, budget scaled into fresh cohorts.

16. Gross Margin % Changes

UNIT PROFITABILITY

$$\text{Gross Margin \%} = (\text{Gross Profit} \div \text{Revenue}) \times 100$$

▼ GROSS MARGIN DECREASED (↓) — CHECK:

- COGS ↑ (supply cost increase, tariffs, higher fulfillment costs).
- Revenue per unit ↓ (deep promotional discounts, markdown cannibalization).
- **Direct Impact:** Gross Margin ↓ directly triggers *Break-even ROAS* ↑.

▲ GROSS MARGIN INCREASED (↑) — CHECK:

- COGS ↓ or price optimization. Allows profitable scaling at lower ROAS targets.

17. Break-even ROAS Changes

BENCHMARK

$$\text{Break-even ROAS} = 1 \div \text{Gross Margin \%}$$

STRATEGIC FRAMEWORK:

- **Gross Margin ↓** → Break-even ROAS ↑ (requires higher efficiency to survive).
- **Gross Margin ↑** → Break-even ROAS ↓ (enables aggressive scaling leeway).
- **Rule:** If Actual ROAS < Break-even ROAS, every ad dollar burns capital.

18. Contribution / Profit

NET PROFITABILITY

$$\text{Profit} = \text{Revenue} - \text{COGS} - \text{Ad Spend} - \text{Variable Costs}$$

▼ PROFIT DECREASED (↓) — CHECK:

- Revenue ↓, COGS ↑, Ad Spend ↑, or Variable Costs ↑ (or combination).
- Check whether scaling ad spend triggered diminishing marginal returns where marginal CPA outpaced marginal gross profit.

▲ PROFIT INCREASED (↑) — CHECK:

- Revenue expanding faster than total marginal COGS and ad spend combined.